

Liénard Systems and Relaxation Oscillations

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Definition: Liénard equation

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. A **Liénard equation** is a second-order equation

$$\ddot{x} + f(x)\dot{x} + g(x) = 0.$$

Recall: Position–velocity form

For the Liénard equation

$$\ddot{x} + f(x)\dot{x} + g(x) = 0,$$

set $v = \dot{x}$. Its position–velocity system is

$$\dot{x} = v, \quad \dot{v} = -g(x) - f(x)v.$$

Notation: Liénard energy function

Write

$$G(x) = \int_0^x g(u) du, \quad H(x, v) = \frac{1}{2}v^2 + G(x).$$

Observation: Liénard energy balance

Along a solution of the position–velocity system,

$$\dot{H} = v\dot{v} + g(x)\dot{x} = -f(x)v^2.$$

Thus $f(x) > 0$ removes energy and $f(x) < 0$ supplies energy whenever $v \neq 0$.

Theorem: Liénard's theorem

Let $f, g \in C^1(\mathbb{R})$. Assume:

1. g is odd and $g(x) > 0$ for every $x > 0$;
2. f is even;
3. the odd function

$$F(x) = \int_0^x f(u) du$$

has exactly one positive zero a ;

4. $F(x) < 0$ for $0 < x < a$;
5. for $x > a$, the function F is positive and nondecreasing, and

$$F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Then

$$\ddot{x} + f(x)\dot{x} + g(x) = 0$$

has exactly one limit cycle surrounding the origin, and that limit cycle is stable.

Question: Apply Liénard's theorem to the van der Pol oscillator

Prove that, for every $\mu > 0$,

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

has exactly one stable limit cycle.

Solution. Here

$$f(x) = \mu(x^2 - 1), \quad g(x) = x.$$

Both functions are continuously differentiable, g is odd and positive for $x > 0$, and f is even. Moreover,

$$\begin{aligned} F(x) &= \int_0^x \mu(u^2 - 1) du \\ &= \mu \left(\frac{x^3}{3} - x \right) \\ &= \frac{\mu x}{3} (x^2 - 3). \end{aligned}$$

The only positive zero is

$$a = \sqrt{3}.$$

For $0 < x < \sqrt{3}$, $F(x) < 0$. For $x > \sqrt{3}$, $F(x) > 0$ and

$$F'(x) = \mu(x^2 - 1) > 0.$$

Also,

$$F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Every hypothesis of Liénard's theorem holds. Hence the van der Pol oscillator has exactly one stable limit cycle for every $\mu > 0$.

Definition: Planar slow-fast system

Let $U \subseteq \mathbb{R}^2$ be open, let $\varepsilon_0 > 0$, and let

$$Q \in C^2(U \times (-\varepsilon_0, \varepsilon_0), \mathbb{R}), \quad P \in C^1(U \times (-\varepsilon_0, \varepsilon_0), \mathbb{R}).$$

A **planar slow-fast system in fast time** is

$$x' = Q(x, y, \varepsilon), \quad y' = \varepsilon P(x, y, \varepsilon), \quad 0 < \varepsilon < \varepsilon_0,$$

where $(x, y) \in U$ and the prime denotes differentiation with respect to the fast time s .

Definition: Critical manifold

For a planar slow-fast system in fast time

$$x' = Q(x, y, \varepsilon), \quad y' = \varepsilon P(x, y, \varepsilon), \quad (x, y) \in U,$$

the **critical manifold** of the singular problem $\varepsilon = 0$ is

$$\mathcal{C}_0 = \{(x, y) \in U : Q(x, y, 0) = 0\}.$$

Definition: Attracting branch

For a scalar fast variable x , a connected C^1 curve $\mathcal{B} \subseteq \mathcal{C}_0$ is an **attracting branch** if

$$Q_x(x, y, 0) < 0$$

for every $(x, y) \in \mathcal{B}$.

Definition: Repelling branch

For a scalar fast variable x , a connected C^1 curve $\mathcal{B} \subseteq \mathcal{C}_0$ is a **repelling branch** if

$$Q_x(x, y, 0) > 0$$

for every $(x, y) \in \mathcal{B}$.

Definition: Fold

A point $(x_*, y_*) \in \mathcal{C}_0$ is a **generic fold**, or **knee**, if

$$Q_x(x_*, y_*, 0) = 0, \quad Q_{xx}(x_*, y_*, 0) \neq 0, \quad Q_y(x_*, y_*, 0) \neq 0.$$

Idea: Relaxation-oscillation geometry

For a planar slow–fast family, a **relaxation oscillation** is a family of periodic motions that, as $\varepsilon \downarrow 0$, spends most of each period near attracting branches of the critical manifold and connects those slow portions by transitions on the fast time scale.

Question: Derive the large-parameter slow–fast coordinates

To study the limit $\mu \rightarrow \infty$, rewrite

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

so that its critical manifold is visible.

Solution. Define

$$F_0(x) = \frac{x^3}{3} - x, \quad y = \frac{\dot{x}}{\mu} + F_0(x).$$

Since

$$F'_0(x) = x^2 - 1,$$

the van der Pol equation gives

$$\dot{y} = \frac{\ddot{x} + \mu(x^2 - 1)\dot{x}}{\mu} = -\frac{x}{\mu}.$$

Rearranging the definition of y gives

$$\boxed{\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}.}$$

With fast time $s = \mu t$ and $\varepsilon = \mu^{-2}$, this becomes

$$x' = y - F_0(x), \quad y' = -\varepsilon x.$$

Here the prime denotes differentiation with respect to s . The critical manifold is

$$\mathcal{C}_0 = \{(x, y) : y = F_0(x)\}.$$

Question: Classify the critical cubic

For

$$\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}, \quad \mu > 0, \quad F_0(x) = \frac{x^3}{3} - x,$$

classify the branches and folds of the critical manifold $y = F_0(x)$.

Solution. With $s = \mu t$ and $\varepsilon = \mu^{-2}$, the fast-time system has

$$Q(x, y, 0) = y - F_0(x), \quad Q_x(x, y, 0) = 1 - x^2.$$

Hence

$$\begin{array}{l|l} |x| > 1 & \text{attracting branch} \\ |x| < 1 & \text{repelling branch} \\ x = \pm 1 & \text{folds} \end{array}$$

Question: Classify the critical cubic (continued)

At $x = \pm 1$,

$$Q_x = 0, \quad Q_{xx} = -2x \neq 0, \quad Q_y = 1 \neq 0,$$

so both folds are generic. Their coordinates are

$$(1, -2/3) \quad \text{and} \quad (-1, 2/3).$$

Definition: Singular relaxation cycle

For this singular slow-fast problem, the **singular relaxation cycle** is the oriented piecewise closed curve consisting of segments on the attracting branches of the critical manifold, traversed in the direction determined by $\dot{y} = -x/\mu$, and horizontal fast jumps between those branches.

Visualization: Geometry of the singular relaxation cycle

The critical cubic is

$$y = F_0(x) = \frac{x^3}{3} - x.$$

Its folds are

$$(1, -2/3) \quad \text{and} \quad (-1, 2/3).$$

Horizontal fast jumps have the limiting endpoints

$$(1, -2/3) \longrightarrow (-2, -2/3),$$

$$(-1, 2/3) \longrightarrow (2, 2/3),$$

because

$$F_0(x) + \frac{2}{3} = \frac{1}{3}(x-1)^2(x+2)$$

and

$$F_0(x) - \frac{2}{3} = \frac{1}{3}(x+1)^2(x-2).$$

Since

$$\dot{y} = -\frac{x}{\mu},$$

the oriented singular cycle is

segment	x -range	direction
right slow branch	$2 \rightarrow 1$	downward
lower fast jump	$1 \rightarrow -2$	left
left slow branch	$-2 \rightarrow -1$	upward
upper fast jump	$-1 \rightarrow 2$	right

The slow segments take time of order μ ; the fast jumps take time of order μ^{-1} away from neighborhoods of the fold points.

Question: Compute the leading relaxation period

Estimate the period $T(\mu)$ of the van der Pol limit cycle as $\mu \rightarrow \infty$.

Solution. Introduce the slow time

$$\tau = \frac{t}{\mu}.$$

In the singular reduced problem,

$$0 = y - F_0(x), \quad \frac{dy}{d\tau} = -x.$$

Thus, along either smooth outer branch,

$$y = F_0(x),$$

and therefore

$$F'_0(x) \frac{dx}{d\tau} = -x.$$

Since $F'_0(x) = x^2 - 1$,

$$\frac{d\tau}{dx} = -\frac{x^2 - 1}{x}.$$

The right slow segment runs from $x = 2$ to $x = 1$, so

$$\begin{aligned} T_{\text{right}}(\mu) &= \mu \int_1^2 \left(x - \frac{1}{x} \right) dx + o(\mu) \\ &= \mu \left[\frac{x^2}{2} - \ln x \right]_1^2 + o(\mu) \\ &= \mu \left(\frac{3}{2} - \ln 2 \right) + o(\mu) \quad (\mu \rightarrow \infty). \end{aligned}$$

By symmetry, the left slow segment has the same leading duration. The fast jumps contribute $o(\mu)$ to the period. Therefore

$$\boxed{T(\mu) = \mu(3 - 2 \ln 2) + o(\mu)} \quad (\mu \rightarrow \infty).$$

Visualization: Conclusions and their hypotheses

conclusion	hypothesis	scope
one stable limit cycle	all Liénard hypotheses	$\mu > 0$
slow outer-branch motion and fast jumps	$\mu \rightarrow \infty$	leading singular geometry
$T(\mu) = \mu(3 - 2 \ln 2) + o(\mu)$	$\mu \rightarrow \infty$	leading period

Observation: Failure of a Liénard hypothesis

Failure of a Liénard hypothesis makes the theorem inconclusive; it does not prove that a limit cycle is absent.

Question: Small parameter in the van der Pol oscillator

For the van der Pol oscillator

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0,$$

suppose $\mu > 0$ is small. Which conclusions remain valid, and which approximation is not justified?

Answer. Liénard's theorem still gives exactly one stable limit cycle. The large- μ relaxation approximation is not justified.

Question: Equilibrium-free trapping annulus

A compact trapping annulus contains no equilibrium. What follows?

Answer. The Poincaré–Bendixson theorem gives at least one periodic orbit, but not uniqueness, isolation, or stability.

Question: Two positive zeros of the Liénard integral

For a Liénard equation, suppose

$$F(x) = \int_0^x f(u) du$$

has two positive zeros. What does Liénard's theorem imply?

Answer. Liénard's theorem is inconclusive because its one-positive-zero hypothesis fails.

Question: Is the critical cubic invariant at a finite parameter?

For fixed $\mu > 0$, is

$$y = F_0(x), \quad F_0(x) = \frac{x^3}{3} - x,$$

an invariant curve of

$$\dot{x} = \mu(y - F_0(x)), \quad \dot{y} = -\frac{x}{\mu}?$$

Solution. Set $h(x, y) = y - F_0(x)$. Along a solution,

$$\begin{aligned} \dot{h} &= \dot{y} - F_0'(x)\dot{x} \\ &= -\frac{x}{\mu} - \mu(x^2 - 1)(y - F_0(x)). \end{aligned}$$

On the cubic $h = 0$, this reduces to $\dot{h} = -x/\mu$. Thus the vector field crosses the cubic at every point with $x \neq 0$. Consequently, for finite μ , the cubic is the exact x -nullcline but is not an invariant curve. It is the critical manifold of the singular problem and approximates the slow portions of the orbit when μ is sufficiently large, away from the folds.